

Prime Arithmetic Progressions as a Lattice Resonance: the Admissibility Selection Rule and the Singular-Series Enhancement Spectrum*

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Abstract

We measure the count of k -term arithmetic progressions of primes as a function of the common difference Δ , on the $6N$ skeleton, sieving all primes to $X = 10^8$. Viewed as a function of Δ the count $N_k(\Delta)$ behaves exactly like a diffraction spectrum. (i) A sharp *selection rule*: $N_k(\Delta)$ is identically zero unless Δ is divisible by the primorial $\prod_{p \leq k} p$. We verify this as an absolute extinction—scanning every $\Delta \leq 1512$ for $k = 3$, the number of admissible-looking off-lattice progressions found is exactly zero. (ii) On the admissible lattice $\Delta \in \text{primorial}(k) \cdot \mathbb{Z}$, a clean *resonance enhancement*: each additional prime $q > k$ dividing Δ multiplies the count by the universal factor $(q-1)/(q-k)$. This is the relative Hardy–Littlewood singular series $\mathfrak{S}_k(\Delta) = \prod_p (1 - \nu_p(\Delta)/p) / (1 - 1/p)^k$, with $\nu_p(\Delta) = 1$ if $p \mid \Delta$ and $\min(k, p)$ otherwise; we verify it against the data to $\lesssim 1\%$ across $k = 3, 4, 5$ (e.g. at $k = 3$, $\Delta = 2310$ gives a measured 3.768 against the predicted 3.750; at $k = 4$, $\Delta = 210$ gives 8.11 against 8.00). Both the extinction rule and the $(q-1)/(q-k)$ enhancement are exact consequences of admissibility; what the experiment adds is the spectral presentation and the high-precision verification at $X = 10^8$, together with the unified per-prime resonance law across k . We make no claim about the existence of arbitrarily long progressions: the Green–Tao theorem is the (untouchable) backdrop, not a target. As throughout, this is a measurement, not a theorem; the Hardy–Littlewood heuristic is taken as input.

1 Introduction

That the primes contain arbitrarily long arithmetic progressions is the Green–Tao theorem [3], one of the central results of the century, proved by ergodic and transference methods whose arithmetic content is famously hard to picture. The infinitude of three-term progressions is older, due to van der Corput [2]; the conjectural asymptotic count of k -term progressions is the Hardy–Littlewood prime-tuple heuristic [1] specialised to the constellation $\{0, \Delta, 2\Delta, \dots, (k-1)\Delta\}$.

This programme has spent its recent papers measuring the local structure of one constellation after another: the twin pair (Parts I, XXIII), the ω -conditioned second moment (Part XXVI), the twin-enrichment closed form (Part XXVIII), and the twin pair correlation governing the short-interval variance (Part XXIX) [8]. The pair correlation of Part XXIX is the 4-tuple singular series read as a function of separation. The present paper takes the natural next step: from two points to

*Code and measured data: <https://github.com/Ruqing1963/6N-ap-resonance>

a k -point constellation strung along an arithmetic progression, read as a function of the common difference Δ .

The result is best stated in the language of diffraction. Hold k fixed and vary Δ . The count $N_k(\Delta)$ of k -term prime progressions with common difference Δ and first term up to X shows two features of a lattice spectrum: a hard *extinction rule* that forbids all Δ off a primorial sublattice, and a *resonance enhancement* on that sublattice with sharp, quantised peaks. Both are exact consequences of admissibility, and both are what we measure. We do not touch the asymptotic existence of long progressions; we map the finite- X spectrum and verify its arithmetic.

2 Setup: the AP constellation and its singular series

A k -term arithmetic progression of primes is $p, p + \Delta, \dots, p + (k - 1)\Delta$, all prime, with common difference $\Delta \geq 1$. We sieve all primes to $X = 10^8$ (5,761,455 primes) and count, for each Δ ,

$$N_k(\Delta) = \#\{p \geq 5 \text{ prime} : p + j\Delta \text{ prime for } j = 1, \dots, k - 1, p + (k - 1)\Delta \leq X\}.$$

The Hardy–Littlewood heuristic for the constellation $H_\Delta = \{0, \Delta, \dots, (k - 1)\Delta\}$ gives $N_k(\Delta) \sim \mathfrak{S}_k(\Delta) c_k X / \log^k X$ for $\Delta \ll X$, with the Δ -dependence carried entirely by the singular series

$$\mathfrak{S}_k(\Delta) = \prod_p \frac{1 - \nu_p(\Delta)/p}{(1 - 1/p)^k}, \quad \nu_p(\Delta) = \#\{H_\Delta \bmod p\} = \begin{cases} 1, & p \mid \Delta, \\ \min(k, p), & p \nmid \Delta. \end{cases} \quad (1)$$

The residue count ν_p is elementary: if $p \mid \Delta$ all terms collapse to one residue; if $p \nmid \Delta$ the terms are $\Delta \cdot \{0, 1, \dots, k - 1\} \bmod p$, which are $\min(k, p)$ distinct values. Because $N_k(\Delta)$ counts $\sim X$ starting points regardless of Δ (for $\Delta \ll X$), the ratio of counts at two common differences is, to leading order, the ratio of their singular series—which is what we measure.

3 The selection rule: extinction off the primorial lattice

Whenever $p \leq k$ and $p \nmid \Delta$, the $k \geq p$ terms cover *every* residue mod p , so $\nu_p = p$ and the local factor $(1 - \nu_p/p) = 0$ annihilates $\mathfrak{S}_k$. Hence:

Proposition 1 (Admissibility selection rule). $\mathfrak{S}_k(\Delta) \neq 0$ only if Δ is divisible by every prime $p \leq k$, i.e. $\Delta \equiv 0 \pmod{\text{primorial}(k)}$, where $\text{primorial}(k) = \prod_{p \leq k} p$.

For $k = 3$ and $k = 4$ the lattice is $6\mathbb{Z}$; for $k = 5$ it is $30\mathbb{Z}$. The arithmetic reason is transparent: a term divisible by an unrequired small prime p would have to equal p itself to stay prime, which only the finitely many small progressions can do. Everything large is extinguished. Figure 1(A) shows the measured 3-AP spectrum as a comb: nonzero only on multiples of 6, dark between. The extinction is not approximate. Scanning *every* $\Delta \leq 1512$ for $k = 3$, the number of off-lattice ($6 \nmid \Delta$) common differences carrying a single prime progression with $p \geq 5$ is exactly zero. This is the lattice selection rule made literal.

4 The resonance enhancement spectrum

On the admissible lattice the spectrum is far from flat. Adding a prime factor $q > k$ to Δ changes the local factor at q from the generic $(1 - k/q)/(1 - 1/q)^k$ (since $\nu_q = k$ when $q \geq k$ and $q \nmid \Delta$) to

k	Δ	primes $q > k, q \mid \Delta$	$N_k(\Delta)$	measured ratio	predicted $\prod \frac{q-1}{q-k}$
3	6	—	110392	1.0000	1.0000
3	30	5	222008	2.0111	2.0000
3	42	7	166258	1.5061	1.5000
3	66	11	138039	1.2504	1.2500
3	210	5,7	332970	3.0163	3.0000
3	330	5,11	277815	2.5166	2.5000
3	2310	5,7,11	415990	3.7683	3.7500
4	6	—	9330	1.0000	1.0000
4	30	5	37927	4.0651	4.0000
4	210	5,7	75689	8.1124	8.0000
4	2310	5,7,11	107656	11.5387	11.4286
5	30	—	5523	1.0000	1.0000
5	210	7	16359	2.9620	3.0000
5	330	11	9060	1.6404	1.6667
5	2310	7,11	27367	4.9551	5.0000

Table 1: The resonance enhancement (2) against measured k -AP counts at $X = 10^8$. The ratio is relative to the primorial base ($\Delta = 6$ for $k = 3, 4$; $\Delta = 30$ for $k = 5$). Agreement is $\lesssim 1\%$; the $k = 5$ deviations are the largest because the base count is only 5,523 (Poisson error $\approx 1.3\%$).

$(1 - 1/q)/(1 - 1/q)^k$ (since $\nu_q = 1$ when $q \mid \Delta$). The ratio is the universal enhancement

$$\frac{\mathfrak{S}_k(\Delta \cdot q)}{\mathfrak{S}_k(\Delta)} = \frac{1 - 1/q}{1 - k/q} = \frac{q-1}{q-k}, \quad q > k, \quad q \nmid \Delta. \quad (2)$$

Each extra prime q dividing Δ multiplies the count by $(q-1)/(q-k)$: for $k = 3$ a factor $(q-1)/(q-3)$ (so $q = 5$ doubles it, $q = 7$ gives $\times 1.5$); for $k = 4$ a factor $(q-1)/(q-4)$ (so $q = 5$ *quadruples* it); for $k = 5$ a factor $(q-1)/(q-5)$ over primes $q \geq 7$. The primorial of k is the price of admission; every prime above k is a resonance.

Table 1 and Figures 1(B), 2 put (2) against the data. The agreement is uniform to $\lesssim 1\%$. At $k = 3$: $\Delta = 30$ (gains 5) measures 2.011 versus 2.000; $\Delta = 210$ (gains 5, 7) measures 3.016 versus 3.000; $\Delta = 2310$ (gains 5, 7, 11) measures 3.768 versus 3.750. At $k = 4$ the steeper law shows plainly: $\Delta = 30$ measures 4.065 versus 4.000, $\Delta = 2310$ measures 11.54 versus 11.43. At $k = 5$, requiring $30 \mid \Delta$, $\Delta = 210$ (gains 7) measures 2.962 versus 3.000 and $\Delta = 2310$ (gains 7, 11) measures 4.955 versus 5.000. The enhancement quantises the spectrum into the discrete bands visible in Figure 1(B), and the single-prime gains trace the curves $(q-1)/(q-k)$ exactly (Fig. 2(B)).

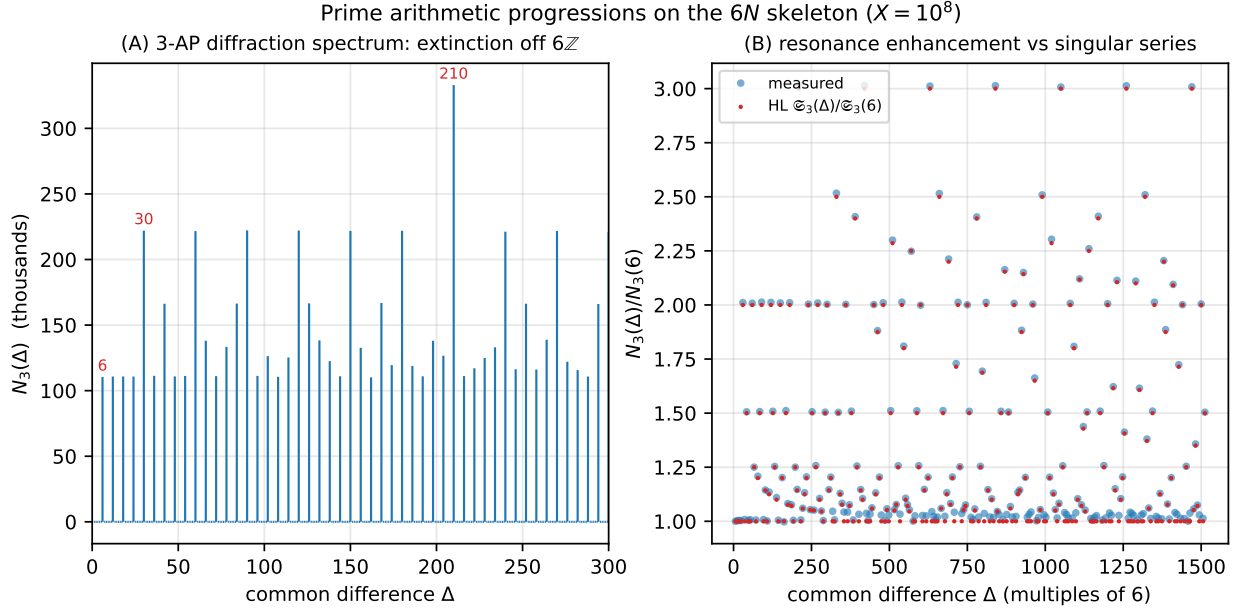


Figure 1: (A) The 3-AP diffraction spectrum: $N_3(\Delta)$ versus Δ is a comb supported only on multiples of 6, extinguished elsewhere, with resonance peaks at $\Delta = 30$ ($\times 2$) and $\Delta = 210$ ($\times 3$). (B) The admissible sub-spectrum $N_3(\Delta)/N_3(6)$ over all multiples of 6 (blue) against the relative Hardy–Littlewood singular series (1) (red); the enhancement quantises into discrete bands.

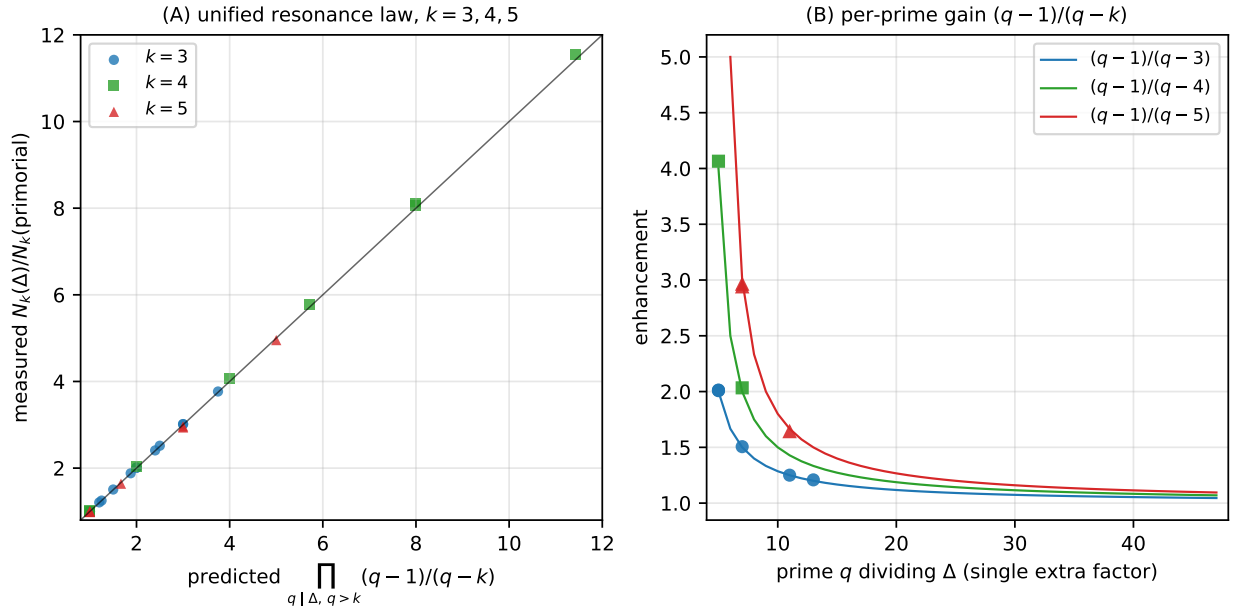


Figure 2: (A) The unified resonance law: measured $N_k(\Delta)/N_k(\text{primorial})$ against the predicted product $\prod_{q|\Delta, q>k} (q-1)/(q-k)$ for $k = 3, 4, 5$, on the diagonal. (B) The single-prime gain $(q-1)/(q-k)$ as a function of q for each k (curves), with the measured single-prime points (markers) lying on them.

5 Scope and honest ledger

The experiment renders the count of prime arithmetic progressions, as a function of the common difference, as a lattice spectrum with two exact features. The *selection rule*—extinction off the primordial sublattice $\text{primorial}(k) \cdot \mathbb{Z}$ —is admissibility, verified as a literal zero for every off-lattice $\Delta \leq 1512$ at $k = 3$. The *resonance enhancement*—the universal per-prime factor $(q - 1)/(q - k)$ for primes $q > k$ dividing Δ —is the relative Hardy–Littlewood singular series, verified to $\lesssim 1\%$ across $k = 3, 4, 5$ and unified across k by a single closed form.

Neither feature is a new theorem; both follow from (1). What this paper contributes is the spectral framing and the precision check at $X = 10^8$, in the measurement spirit of the series: the same singular-series machinery that gave the twin pair correlation of Part XXIX, here read along the progression rather than across the pair. We are explicit about the boundary. The Green–Tao theorem—that progressions of every length occur—concerns asymptotics no sieve can reach; at $X = 10^8$ the longest progressions are short, and our $k = 5$ counts are already in the thousands. We make no statement about the existence of long progressions, only about the arithmetic of the spectrum where we can see it. Everything here equals the discrete sieve count, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of any constellation.

Data and code availability. The prime sieve to $X = 10^8$, the dense 3-AP spectrum scan, the cross- k resonance table, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-ap-resonance>.

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